

[ReadMe: monte_carlo_gps_constraint.ipynb](#)

Step 1 — Imports and Paths: All required libraries (GeoPandas, NumPy, Matplotlib) and file paths are defined using relative paths so the notebook runs for any collaborator who clones the repository without editing paths. Key constants are defined here including `N_SIMULATIONS = 1000`, `RURAL_DISPLACEMENT = 5000m`, `URBAN_DISPLACEMENT = 2000m`, and `PILOT_DISTRICT = "099"` (Ganganagar, Rajasthan). This notebook is the revised version of `monte_carlo_gp_assignment.ipynb`, incorporating the DHS admin2 district constraint using official SHRUG PC11 district boundary polygons. The pipeline is first validated on Ganganagar before scaling up to all of Rajasthan and UP.

Step 2 — Load SHRUG District Polygons: Official SHRUG PC11 district boundary polygons are loaded for all of India (641 districts) and reprojected to UTM Zone 44N. These are used to enforce the DHS displacement constraint — per the DHS GPS displacement README, cluster displacement is restricted to stay within the same admin2 (district) area. Using official district polygons rather than dissolving village polygons avoids gaps and inaccuracies at district edges from imperfect village polygon tiling.

Step 3 — Load SHRUG-LGD Crosswalk: The SHRUG-to-LGD crosswalk is loaded and GP LGD codes are assigned to each village using the same `shrid2` composite key approach as in `cluster_GPs_cleaning.ipynb`.

Step 4 — Load and Prepare Ganganagar Village Polygons: Rajasthan village polygons are loaded and filtered to Ganganagar district (`pc11_d_id = "099"`), yielding 3,034 village polygons, reprojected to UTM Zone 44N.

Step 5 — Dissolve to GP Polygons and Get Ganganagar District Polygon: Village polygons are dissolved to 343 GP-level boundary polygons for Ganganagar. The Ganganagar district polygon is extracted from the official SHRUG district file for use as the displacement constraint.

Step 6 — Load NFHS Clusters for Ganganagar: NFHS4 and NFHS5 clusters are loaded and filtered to Ganganagar using the `DHSREGNA` district name field. NFHS4 yields 43 clusters (31 rural) and NFHS5 yields 45 clusters (33 rural).

Step 7 — Monte Carlo GP Assignment Function with District Constraint: Defines `run_monte_carlo_constrained` — the core function used throughout this notebook. For each cluster, 1,000 random displacements are simulated using `angle ~ Uniform[0, 2π)` and `distance ~ Uniform[0, R]`. Simulated points outside the cluster's district are discarded using `sjoin_nearest` to assign each cluster to its nearest district polygon (handling edge cases where the displaced GPS falls just outside the SHRUG boundary due to source mismatch). `P(primary GP)` is normalised by `n_in_district` — the number of valid draws within the district — rather than total simulations.

Step 8 — Run Monte Carlo for Ganganagar: The constrained Monte Carlo is run for both NFHS4 (31 rural clusters) and NFHS5 (33 rural clusters) in Ganganagar.

Step 9 — Summary Statistics: Key statistics are computed for rural clusters including mean and median `P(primary GP)`, share above probability thresholds, mean GPs hit, mean simulations outside district, and `P(primary GP)` distribution across bins.

Step 10 — District-Level Visualisation: Side-by-side map of Ganganagar for NFHS4 and NFHS5 with clusters coloured by `P(primary GP)` and district boundary shown in blue dashed line.

Step 11 — Zoomed-In Cluster Detail Maps: Individual zoomed-in maps for three hand-check clusters — high (`P = 71.3%`), medium (`P = 36.3%`), and low/border (`P = 61.0%`). Each map shows the cluster GPS point, 5km displacement radius, district boundary, and GP boundaries labelled with probabilities. The border cluster improved dramatically from `P = 24.9%` to `P = 61.0%` after applying the district constraint, confirming it is working correctly. A boundary mismatch cluster (cluster 9924, NFHS5) was identified where the displaced GPS falls outside the SHRUG district polygon due to a boundary source discrepancy between DHS admin2 and SHRUG 2011 Census boundaries — flagged as a known limitation affecting a small number of edge cases.

Step 12 — Save Ganganagar Results: Results saved to `../outputs/monte carlo simulation ganganagar/` as `ganganagar_mc_nfhs4_constrained.csv` and `ganganagar_mc_nfhs5_constrained.csv`.

Step 13 — Load All Rajasthan Village Polygons and Assign GP Codes: All 44,976 Rajasthan village polygons are loaded and GP codes assigned. GP coverage is 87.6% (39,420 villages), yielding 11,107 unique GPs across 33 districts.

Step 14 — Dissolve All Rajasthan Villages to GP Polygons and Load District Polygons: Village polygons are dissolved to 11,107 GP boundary polygons for all of Rajasthan, saved to `../outputs/monte carlo simulation rajasthan/rajasthan_gp_polygons_constrained.gpkg`. All 33 Rajasthan district polygons are loaded from the official SHRUG district file.

Step 15 — Load All Rajasthan NFHS Clusters: Rural NFHS4 and NFHS5 clusters loaded and filtered to Rajasthan, yielding 1,190 (NFHS4) and 1,163 (NFHS5) rural clusters across all 33 districts.

Step 16 — Run Monte Carlo for All Rajasthan Clusters: The constrained Monte Carlo is run across all Rajasthan rural clusters. Zero boundary mismatch clusters in either round.

Step 17 — Summary Statistics and District Breakdown (Rajasthan): Full summary statistics and district-level breakdown of mean `P(primary GP)`.

Step 18 — Visualisation and Save (Rajasthan): Map and results saved to `../outputs/monte carlo simulation rajasthan/` with `_constrained` suffix.

Step 19 — Load All UP Village Polygons and Assign GP Codes: UP village polygons loaded from two files (split due to GitHub's 100MB file size limit), concatenated to 107,709 villages. GP coverage is 52.0% (56,051 villages), yielding 33,854 unique GPs — substantially lower than Rajasthan due to the known ~40% LGD data coverage gap.

Step 20 — Dissolve All UP Villages to GP Polygons and Load District Polygons: Village polygons dissolved to 33,854 GP boundary polygons, saved to `../outputs/monte carlo simulation uttar pradesh/up_gp_polygons_constrained.gpkg`. All 71 UP district polygons loaded from the official SHRUG district file.

Step 21 — Load All UP NFHS Clusters: Rural NFHS4 and NFHS5 clusters loaded and filtered to UP, yielding 2,659 (NFHS4) and 2,661 (NFHS5) rural clusters across 71 and 75 districts respectively.

Step 22 — Run Monte Carlo for All UP Clusters: The constrained Monte Carlo is run across all UP rural clusters. Zero boundary mismatch clusters in either round.

Step 23 — Summary Statistics, Visualisation and Save (UP): Full summary statistics, district-level breakdown, map, and results saved to `../outputs/monte carlo simulation uttar pradesh/` with `_constrained` suffix.

Key Outputs and Takeaways:

Ganganagar Pilot: The constrained Monte Carlo confirms and quantifies the findings from `cluster_GPs_cleaning.ipynb`. For rural Ganganagar clusters, mean `P(primary GP)` is 0.417 (NFHS4) and 0.393 (NFHS5) — an improvement over the unconstrained run (0.379/0.370). The district constraint particularly improved the border cluster from `P = 24.9%` to `P = 61.0%` by correctly discarding simulations crossing into Hanumangarh. Visual hand-checking confirmed the code is working correctly.

Full Rajasthan:

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	NFHS4 (2015-16)	NFHS5 (2019-21)
Rural clusters	1,190	1,163
Mean P(primary GP)	0.436	0.423
Clusters P > 0.8	40 (3.4%)	34 (2.9%)
Mean GPs hit	8.18	8.36
Mean sims outside district	51.8	44.7
Boundary mismatch clusters	0	0

The district constraint improved mean **P(primary GP)** by ~3 percentage points and doubled the share of clusters with **P > 0.8** compared to the unconstrained run. The geographic pattern persists: Jaisalmer (0.702), Bikaner (0.564), Churu (0.558) highest; Dungarpur (0.282), Banswara (0.291), Bharatpur (0.307) lowest.

Full Uttar Pradesh:

	NFHS4 (2015-16)	NFHS5 (2019-21)
Rural clusters	2,659	2,661
Mean P(primary GP)	0.130	0.125
Clusters P > 0.8	2 (0.1%)	4 (0.2%)
Mean sims outside district	74.4	55.9
Mean sims outside any GP	426.1	451.6
Boundary mismatch clusters	0	0

The district constraint provides a modest improvement over the unconstrained run (0.121→0.130 NFHS4, 0.117→0.125 NFHS5) but UP results remain dominated by the LGD data coverage gap rather than genuine displacement uncertainty. Approximately 43-45% of simulated points fall outside any GP polygon per cluster. Roughly half of all UP districts have mean **P(primary GP)** near zero. Bundelkhand districts retain the highest confidence: Hamirpur (0.346), Banda (0.336), Chitrakoot (0.313).

Pending next steps: Once GP election reservation status data is cleanly merged, the pipeline will be extended to compute **P(correct treatment status)** — the operationally meaningful quantity. A cluster touching multiple GPs is only a problem if those GPs have different reservation statuses.

Known limitation: A small number of clusters have displaced GPS coordinates that fall just outside the SHRUG district polygon boundary due to source mismatch between DHS admin2 definitions and SHRUG 2011 Census boundaries. These are handled via `sjoin_nearest` assignment to the nearest district. One such cluster (cluster 9924, NFHS5 Ganganagar) had all simulated points outside the district polygon and returned $P = 0.0\%$. This affects a very small number of edge cases and does not materially affect the overall results.